B.TECH CSE PROJECT · OCTOBER 2026

Smart Krishi — Precision Agriculture Platform

Rajasthan Technical University · Arya College of Engineering & I.T.
Academic Year 2026–27

Guide Er. Ram Babu Buri · Team Leader Ravina and team members

CONTEXT

Farmers Make Critical Decisions Without Verified, Location-Specific Data

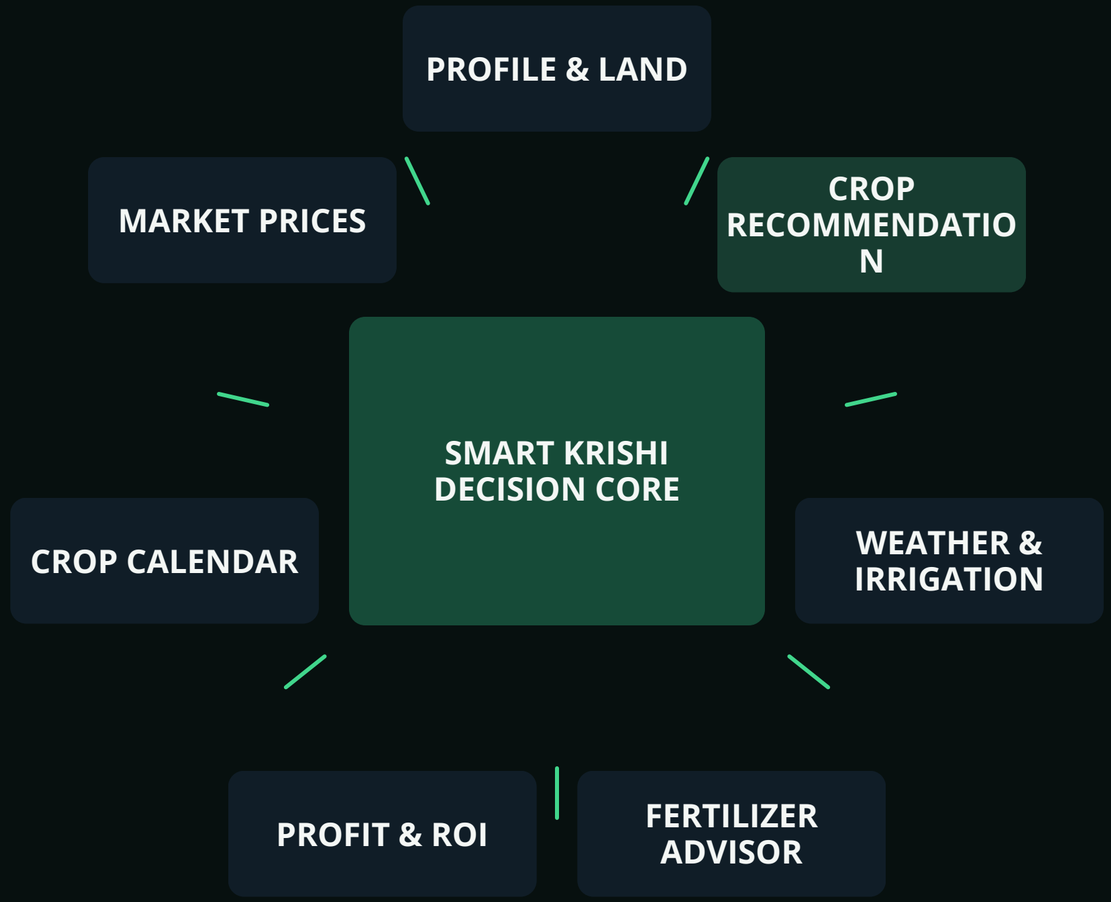
Crop, irrigation, fertilizer, cost and market choices often depend on fragmented or unverified information.

The gap: one trusted, integrated precision-agriculture tool.



One Integrated Decision Support Loop

A responsive web platform connecting farm context to agronomic and market decisions.



Smart Krishi Modules at a Glance

IMPLEMENTED

Farmer & land profiling

Registration, authentication and role-based access

IMPLEMENTED

Precision crop recommendation

10 major Indian crops using soil and climate rules

IMPLEMENTED

Weather & irrigation

Live Open-Meteo telemetry and actionable advisories

IMPLEMENTED

Fertilizer advisor

ICAR-aligned deficit, bag conversion and costing

From Economics to Market Access

IMPLEMENTED

Profit & ROI

Seven cost heads, net profit and break-even

IMPLEMENTED

Growth calendar

Six stages with tasks and advisories

IMPLEMENTED

Price trends

SMA, momentum and confidence — no AI

IMPLEMENTED

Mandi locator

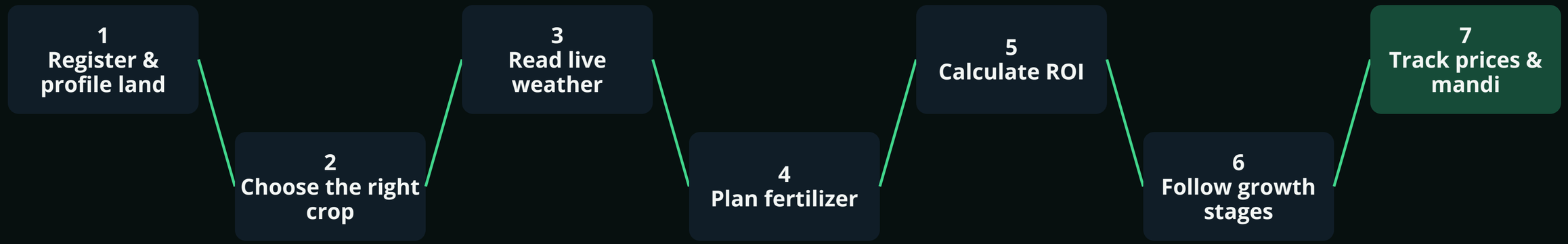
Nearby APMC prices with source tags

IMPLEMENTED

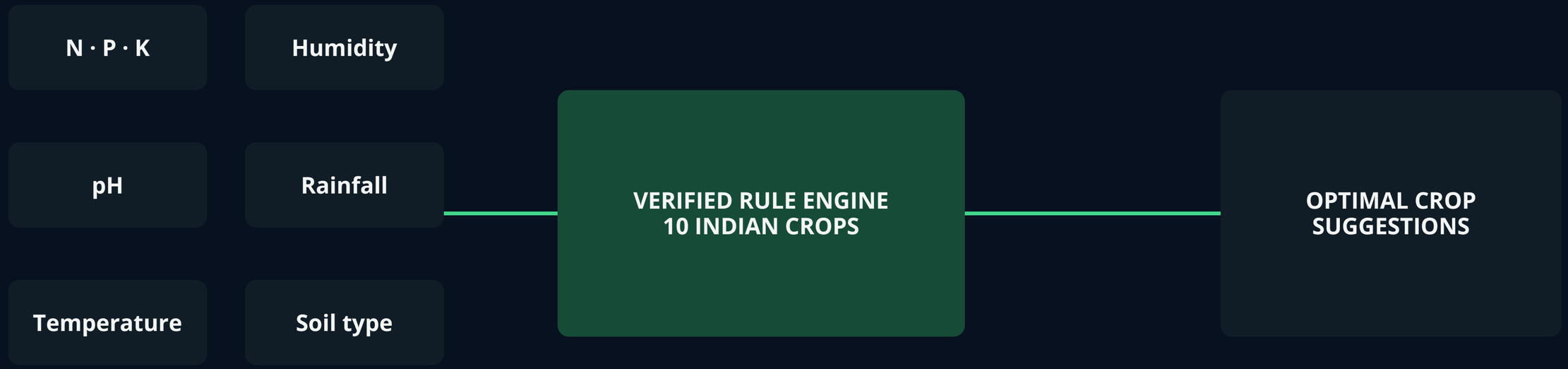
Admin portal

Statistics, prices, schemes and governance

How a Farmer Uses Smart Krishi



Precision Crop Recommendation Logic



Implemented with verified agronomic rules — no AI dependency.

Live Meteorological Data Powers Irrigation Decisions

OPEN-METEO

Geolocation · current conditions · 5-day forecast
Explicit 5,000 ms timeout · zero fake data

PAUSE
Rain detected

HEAVY
High crop need

MODERATE
Balanced demand

LIGHT / SKIP
Low demand

ICAR-Aligned Fertilizer Guidance and Costing

NUTRIENT NEED	COMMERCIAL SOURCE	SCHEDULE
N deficit	Urea / DAP	Split application
P deficit	DAP / SSP	Basal dose
K deficit	MOP	Stage aligned
Micronutrients	Zinc sulphate	Need based

Bag conversion + approximate cost estimation included

Crop Profit and Return on Investment Analysis

NET PROFIT = GROSS REVENUE − 7 COST HEADS

ROI = NET PROFIT ÷ TOTAL COST × 100

Transparent assumptions turn a crop plan into a financial decision.

7
cost heads

₹ / acre
profit view

Break-even
price

Health
classification

Growth Stages & Market Price Trends

1 Sowing

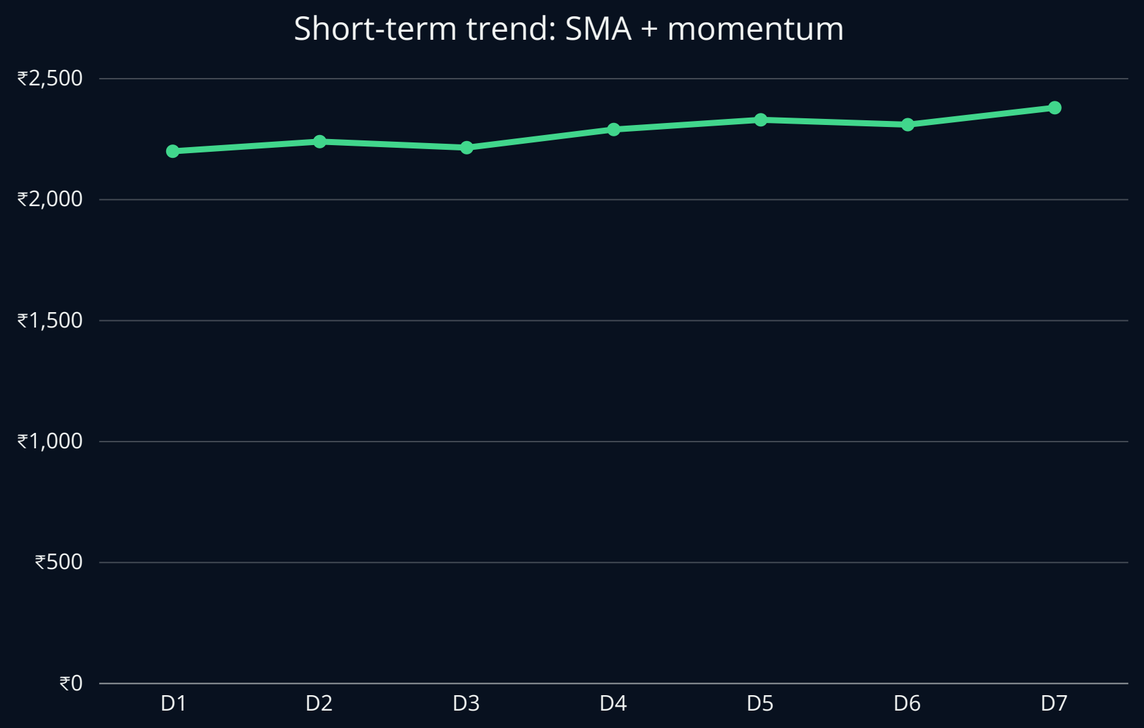
2 Establishment

3 Vegetative

4 Flowering

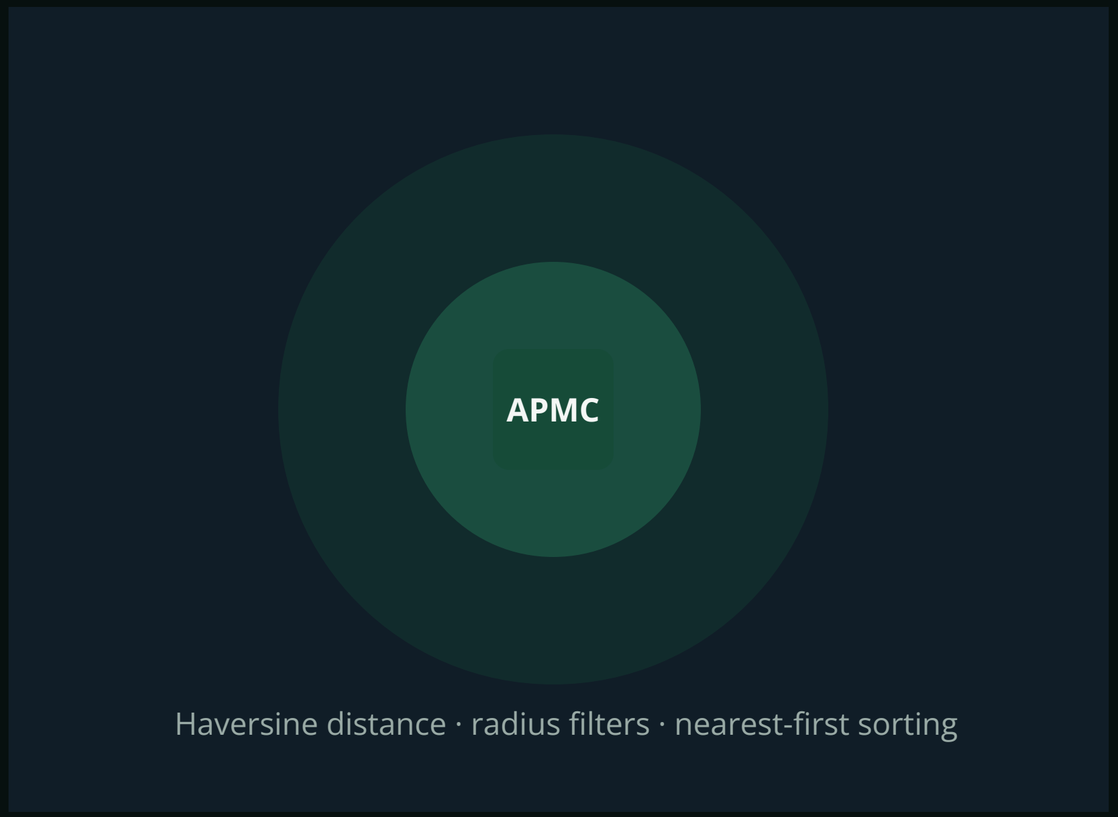
5 Yield formation

6 Harvest



Confidence score communicates uncertainty. No AI claim.

Market Access and Platform Governance



Haversine distance · radius filters · nearest-first sorting

ADMINISTRATION PORTAL

Platform statistics
Farmer directory
Price seeding
Scheme publication

Every price is tagged live, cached or verified fallback.

Distributed Multi-Tier Architecture & Technology Stack

PRESENTATION
HTML5 · CSS3 · ES6+ · Bootstrap 5.3

APPLICATION
Spring Boot 3.3.4 · Java 17 · Security · JWT · JPA

PERSISTENCE
H2 development · MySQL 8.0+ production

EXTERNAL SERVICES
Open-Meteo · data.gov.in / AGMARKNET · geolocation

Dockerized deployment · Netlify frontend · Render backend

Design Constraints for Trustworthy Agriculture Support

01	**Zero fake weather data**	Transparent sourcing only
02	**ICAR-grade adherence**	Verified fertilizer logic
03	**Graceful failure**	Timeouts and explicit fallbacks
04	**No disease AI claim**	Agronomic rules over theatre
05	**Decision support only**	Not transactions or a soil lab

Robust Verification with Zero Failures

16

core JUnit 5 / MockMvc integration tests

0 failures · 0 errors

CRITICAL PATHS COVERED

Authentication & admin access
Crop rules & weather
Fertilizer safety & profit calculation
Calendar & price prediction
Mandi filtering & fallback transparency

Targets: <250 ms calculations · 5,000 ms API timeout · <1.5 s frontend load
Performance benchmarks pending.

Limitations and Roadmap Ahead

CURRENT LIMITS

No AI disease detection
Performance benchmarks pending
Legacy scanner decommissioned

PROPOSED

Enhanced ML price forecasting
Offline mobile support
Soil-lab integration
Farmer feedback loop
Continued scale & UX improvements

Thank You for Your Attention

Questions & Discussion

Ravina · Team Leader, Smart Krishi